

KAI WANG | 王凯

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RESEARCH

- **Galaxy Formation Model:** *Toward a simulation-motivated and observation-calibrated semi-analytical model.*
- **Galaxy Evolution:** *What drives galaxy quenching, metal enrichment, and morphology transformation?*
- **Galaxy-Halo Connection:** *To what extent are the properties of galaxies shaped by their dark matter halos?*
- **Dark Matter Halo:** *How do halo structure, assembly history, and spatial distribution interplay with each other?*

PUBLICATION SUMMARY

◆38 publications; 16 as the first/corresponding author; >469 citations; H-index: >14; open in [NASA/ADS](#)

EXPERIENCE

- Postdoctoral Research Associate, ICC and CEA, **Durham University** Since Jul. 2024
- KIAA Fellow, Kavli Institute for Astronomy and Astrophysics, **Peking University** Jul. 2022 - Jun. 2024

EDUCATION

- Ph.D. in Astronomy, **Tsinghua University** Sep. 2017 - Jul. 2022
Supervisors: Cheng Li & Houjun Mo, Thesis: Finding galaxy groups/clusters at $z\sim 1$ and its application
- Visiting Scholar, **University of Massachusetts, Amherst** Nov. 2019 - Oct. 2021
Supervisor: Houjun Mo
- B.S. in Astronomy, **University of Science and Technology of China (USTC)** Sep. 2013 - Jul. 2017

GRANTS

- KIAA fellow start-up research funding Jul. 2022 - Jul. 2024
(50,000CNY)
- China Scholarship for the Visiting Scholar Nov. 2019 - Oct. 2021
China Scholarship Council (\$45,600)
- National Astronomy Training Base Jun. 2016 - Jun. 2017
Measure the conditional luminosity functions of galaxies at $z\sim 0.6$ using CLAUDS and BOSS (20,000CNY)
- National Astronomy Training Base May 2015 - May 2016
Thermal gravitational-wave background in the general pre-inflationary scenario (20,000CNY)

MENTORSHIP/SUPERVISION

- **Oscar Hickman**, Postgraduate student at Durham University since Feb. 2026
Understanding the distribution of galaxies with semi-analytic models
- **Thomas Power**, L4 student at Durham University since Oct. 2024
The Dance of Milky Way and Andromeda: A Celestial Ballet Across Cosmologies

- **Thomas Richardson**, L4 student at Durham University since Oct. 2024
Where Shadows Follow Light: Baryon-Induced Dark Matter Halo Contraction in Hydrodynamical Simulations
- **Zeyu Gao**, graduate at Peking University Nov. 2022 - Jul. 2024
Decoding the SEDs of galaxies with a prior from hydrodynamical simulations [[arXiv: 2408.07749](#)]
- **Chengyu Ma**, graduate at USTC Dec. 2023 - Jul. 2024
Revisiting the fundamental metallicity relation with observation and simulation [[arXiv: 2407.21716](#)]
- **Haochen Jiang**, undergraduate at USTC Dec. 2023 - Oct. 2025
Dissecting the quenching of massive central galaxies in TNG [[arXiv: 2510.24886](#)]
- **Xunda Sun**, graduate at the University of Chinese Academy of Sciences Jun. 2023 - Oct. 2024
Characterizing the spatial distribution of the metal content for galaxies in FIRE2 [[arXiv: 2409.09290](#)]
- **Zhijun Zhang**, undergraduate at Peking University Sep. 2022 - Jun. 2023
Identify protoclusters from high-redshift photometric surveys [[Bachelor Thesis](#)]

TEACHING

- ExGal Summer School on Astrophysical Simulations Tutorial assistant, Durham University, 2025
- Level-1 Physics Tutorial Tutor, Durham University, 2024
- Cosmology and Galaxy Evolution Guest Lecturer, Peking University, Autumn 2023
- Observational Cosmology Teaching Assistant, Tsinghua University, Autumn 2017
- Particle Cosmology Teaching Assistant, USTC, Spring 2017
- General Relativity Teaching Assistant, USTC, Autumn 2016

PUBLICATION

◆ **38** publications; **16** as the first/corresponding author; **>469** citations; *H*-index: **>14**; open in [NASA/ADS](#)

First and Corresponding[†] author papers:

1. Kai Wang[†], Peder Norberg, Azadeh Fattahi, Louis E. Strigari submitted (2026) [[arXiv: 2601.08419](#)]
[Local Group analogues in a cosmological context -- I. Relating velocity structure to the cosmic web](#)
2. Kai Wang[†], Joop Schaye, Alejandro Benítez-Llambay, Evgenii Chaikin, Carlos S. Frenk, Filip Huško, Robert J. McGibbon, Sylvia Ploeckinger, Alexander J. Richings, Matthieu Schaller, James W. Trayford
MNRAS Letter 546, 1 (2026) [[arXiv: 2510.02573](#)]
[Gravitational potential drives the concentration dependence of the stellar mass-halo mass relation](#)
3. Kai Wang[†], *MNRAS* 545, 1 (2026) [[arXiv: 2510.02573](#)]
[The origin of the galaxy size-stellar metallicity relation: A semi-analytical perspective](#)
4. Haochen Jiang, Enci Wang[†], Kai Wang[†], Chengyu Ma, Xu Kong[†], *ApJ* 994 220 (2025) [[arXiv: 2510.24886](#)]
[Dissecting the mass quenching in TNG50: Galaxy size determines the quenching mode](#)
5. Kai Wang[†], Yingjie Peng[†], *ApJ* 980 233 (2025) [[arXiv: 2408.07743](#)]
[Testing galaxy formation models with the stellar mass-halo mass relations for star-forming and quiescent galaxies](#)
6. Chengyu Ma, Kai Wang[†], Enci Wang[†], Yingjie Peng, Haochen Jiang, Haoran Yu, Cheng Jia, Zeyu Chen, Haixin Li, Xu Kong *ApJL* 971 L14 (2024) [[arXiv: 2407.21716](#)]
[Revisiting the fundamental metallicity relation with observation and simulation](#)

7. Kai Wang[†], Houjun Mo, Yangyao Chen, Huiyuan Wang, Xiaohu Yang, Jiaqi Wang, Yingjie Peng, Zheng Cai *MNRAS* 528, 2046 (2024) [arXiv: 2309.01039]
[Characterizing the assembly of dark matter halos with protohalo size histories: I. Redshift evolution, relation to descendant halos, and halo assembly bias](#)
8. Kai Wang[†], Houjun Mo, Yangyao Chen, Joop Schaye *MNRAS* 527, 10760 (2023) [arXiv: 2310.00200]
[An efficient and robust method to estimate halo concentration based on the method of moments](#)
9. Kai Wang[†], Xin Wang[†], Yangyao Chen, *ApJ* 951, 66 (2023) [arXiv: 2305.08161]
[Environmental dependence of the mass-metallicity relation in cosmological hydrodynamical simulations](#)
10. Kai Wang[†], Yangyao Chen, Qingyang Li, Xiaohu Yang, *MNRAS* 522, 3188 (2023) [arXiv: 2304.07189]
[Late-formed halos prefer to host quiescent central galaxies. I. Observational results](#)
11. Kai Wang[†], Yingjie Peng[†], Yangyao Chen, *MNRAS* 523, 1268 (2023) [arXiv: 2304.06886]
[Dissect two-halo galactic conformity effect: The dependence of star formation activities on the large-scale environment for central galaxies](#)
12. Kai Wang[†], Houjun Mo, Cheng Li, Yangyao Chen, *MNRAS* 520, 1774 (2023) [arXiv: 2211.00485]
[Relating galaxies across different redshift to study galaxy evolution](#)
13. Kai Wang[†], Houjun Mo, Cheng Li, Yangyao Chen, *MNRAS* 505, 3892 (2021) [arXiv: 2104.12223]
[Finding proto-clusters to trace galaxy evolution: I. The finder and its performance](#)
14. Kai Wang[†], Houjun Mo, Cheng Li, Jiacheng Meng, Yangyao Chen, *MNRAS* 499, 89 (2020) [arXiv: 2006.05426]
[Identifying galaxy groups at high redshift from incomplete spectroscopic data: I. The group finder and application to zCOSMOS](#)
15. Kai Wang, Larissa Santos, Jun-Qing Xia, Wen Zhao[†], *JCAP* 01, 053 (2017) [arXiv: 1608.04189]
[Thermal gravitational-wave background in the general pre-inflationary scenario](#)
16. Yi-Fan Wang, Kai Wang[†], Wen Zhao, *RAA* 16, 4 (2016) [arXiv: 1511.01220]
[Smoothing methods comparison for CMB E- and B-mode separation](#)

Co-author papers:

17. Tiancheng Yang et al. *ApJL* 1000 2 (2025) [arXiv: 2510.12235]
[A census of quiescent galaxies across \$0.5 < z < 8\$ with JWST/MIRI: Mass-dependent number density evolution of quiescent galaxies in the early Universe](#)
18. Hanwen Sun et al. *ApJL* 993 2 (2025) [arXiv: 2508.21356]
[The Bigfoot: A footprint of a Coma cluster progenitor at \$z=3.98\$](#)
19. Cheng Jia et al. *ApJL* 986 L24 (2025) [arXiv: 2504.18820]
[Potential-Driven Metal Cycling: JADES Census of Gas-Phase Metallicity for galaxies at \$1 < z < 7\$](#)
20. Xunda Sun et al. *ApJ* 986 179 (2025) [arXiv: 2409.09290]
[The physical origin of positive metallicity radial gradients in high-redshift galaxies: insights from the FIRE-2 cosmological hydrodynamic simulations](#)
21. Dingyi Zhao et al. *ApJ* 979 42 (2025) [arXiv: 2408.12442]
[From Halos to Galaxies. VI. Improved Halo Mass Estimation for SDSS Groups and Measurement of the Halo Mass Function](#)
22. Cheqiu Lyu et al. *ApJ* 972 108 (2025) [arXiv: 2407.03409]
[From Halos to Galaxies. IX. Estimate of Halo Assembly History for SDSS Galaxy Groups](#)
23. Zeyu Gao, Yingjie Peng[†], Kai Wang et al. *ApJ* 979 66 (2024) [arXiv: 2408.07749]
[From Halos to Galaxies. X: Decoding Galaxy SEDs with Physical Priors and Accurate Star Formation History Reconstruction](#)

24. Cheqiu Lyu et al. **ApJ** 959 5 (2024) [arXiv: 2407.03409]
[From Halos to Galaxies. IX. Accurate estimate of halo assembly history for SDSS galaxy groups](#)
25. Qinxun Li et al. **ApJL** 969 L25 (2024) [arXiv: 2402.10740]
[Black-Hole-to-Halo Mass Relation From UNIONS Weak Lensing](#)
26. Tao Wang[†] et al. **Nature** (2023) [arXiv: 2311.07653]
[Black holes regulate cold gas accretion in massive galaxies](#)
27. Yangyao Chen[†], H.J Mo, **Kai Wang**, **MNRAS** 526 2542 (2023) [arXiv: 2304.13890]
[Massive Dark Matter Halos at High Redshift: Implications for Observations in the JWST Era](#)
28. Cheqiu Lyu et al. **ApJ** 959 5 (2023) [arXiv: 2310.10733]
[From Halos to Galaxies. VII. The Connections Between Stellar Mass Growth History, Quenching History, and Halo Assembly History for Central Galaxies](#)
29. Jiacheng Meng et al. **ApJ** 964 2 (2024) [arXiv: 2008.13733]
[Measuring galaxy abundance and clustering at high redshift from incomplete spectroscopic data: Tests on mock catalogs](#)
30. Yangyao Chen et al. **MNRAS** 525 1254 (2023) [arXiv: 2301.08972]
[A Conditional Abundance Matching Method of Extending Simulated Halo Merger Trees to Resolve Low-Mass Progenitors and Sub-halos](#)
31. Qingyang Li et al. **ApJ** 933 9 (2022) [arXiv: 2205.05517]
[Groups and Protocluster Candidates in the CLAUDS and HSC-SSP Joint Deep Surveys](#)
32. Yangyao Chen et al. **MNRAS** 507 2510 (2021) [arXiv: 2106.03984]
[MAHGIC: A Model Adapter for the Halo-Galaxy Inter-Connection](#)
33. Zhaoyu Wang et al. **Sci. China Phys. Mech. Astron.** 64 289811 (2021) [arXiv: 2106.14159]
[The clustering of galaxies in the DESI imaging legacy surveys DR8:I. the luminosity and color dependent intrinsic clustering](#)
34. Yangyao Chen et al. **MNRAS** 504 4865 (2021) [arXiv: 2009.12467]
[How to empirically model star formation in dark matter halos: I. Inferences about central galaxies from numerical simulations](#)
35. Yangyao Chen et al. **ApJ** 899 81 (2020) [arXiv: 2003.05137]
[Relating the structure of dark matter halos to their assembly and environment](#)
36. Jia-Ni Ye, **Kai Wang**, Yi-Fu Cai, **Eur. Phys. J. C** 77:720 (2017) [arXiv: 1705.10956]
[Superconducting cosmic strings as sources of cosmological fast radio bursts](#)
37. Larissa Santo et al. **JCAP** 01 043 (2017) [arXiv: 1612.03564]
[Statistical imprints of CMB B-type polarization leakage in an incomplete sky survey analysis](#)
38. Larissa Santo et al. **JCAP** 07 029 (2016) [arXiv: 1510.07779]
[Probing the statistical properties of CMB \$B\$ -mode polarization through Minkowski Functionals](#)

SERVICE

- **Professional Service**

Referee for MNRAS (since 2024), ApJ (since 2023), and A&A (since 2022)

- **Departmental Service**

SOC member of session "Galaxy chemistries across cosmic time" of NAM 2026	2026
Co-organiser of CosmICConnection at Durham University	2026-
Co-organiser of Friday Lunchtime Astronomy Talks (FLAT) at Durham University	2025-2026

Co-lead of morning arXiv journal club at Durham University	2025
LOC member of National Astronomy Meeting (NAM) 2025 at Durham University	2025
SOC member of session "The Golden Era of Gravitational Lensing" of NAM 2025	2025
Faculty Candidate Interview Committee at KIAA, Postdoc Representative	2023, 2024
Co-organiser of weekly Galaxy Party at KIAA	2023
Co-organiser of the Postdoc Science Day at KIAA	2022
Co-organiser of the Speaker Lunch at the Tsinghua Center for Astrophysics	2018-2019

SEMINAR TALKS

- Seminar of Galaxy and Cosmology at National Astronomical Observatories, CAS (**invited**)
Forged in quenching: An EAGLE's view of morphological transformation Beijing, Apr. 2026
- KIAA-DoA Seminar at Peking University (**invited**)
Forged in quenching: An EAGLE's view of morphological transformation Beijing, Apr. 2026
- Informal Talk at Tsinghua University (**invited**)
Metallicity as an Astrophysical Diagnostic Beijing, Apr. 2026
- Astronomy Seminar, Nanjing University (**invited**)
Forged in quenching: An EAGLE's view of morphological transformation Nanjing, Apr. 2026
- Astronomy Seminar, Westlale University (**invited**)
Revisiting the Classic: How well do we understand galaxy formation before JWST challenges it? Hangzhou, Apr. 2026
- Astrophysics Research Institute Seminar, Liverpool John Moores University (**invited**)
Metallicity as an Astrophysical Diagnostic Liverpool, Nov. 2025
- Lunchtime Talks at Astronomy Group, University of St Andrews (**invited**)
Metallicity as an Astrophysical Diagnostic St Andrews, Oct. 2025
- Informal Talk at Tsinghua University (**invited**)
Local Group Analogs in a cosmological context Beijing, Jun. 2025
- KIAA-DoA Seminar at Peking University (**invited**)
Towards the next-generation semi-analytical galaxy formation model Beijing, Jun. 2025
- USTC Astronomy Seminar Series (**invited**)
Towards the next-generation semi-analytical galaxy formation model Hefei, Jun. 2025
- DoA Seminar at Shanghai Jiao Tong University (**invited**)
Towards the next-generation semi-analytical galaxy formation model Shanghai, May 2025
- Friday Lunchtime Astronomy Talks at Durham University
Local Group Analogs in a cosmological context Durham, Mar. 2025
- Galaxy & Cosmology seminar at Tsinghua University (**invited**)
Dark matter halo and its structure, assembly, and clustering Beijing, May 2024
- Lunch talk at South-Western Institute For Astronomy Research, Yunnan University (**invited**)
Galaxy formation within and without dark matter halos Kunming, Apr. 2024
- Astronomy Seminar at the Huazhong University of Science and Technology (**invited**)
Galaxy formation within and without dark matter halos Wuhan, Mar. 2024
- ITC Luncheon at the Center for Astrophysics I Harvard & Smithsonian (**invited**)
How to connect galaxies across cosmic time? Cambridge, Jan. 2024
- Steward/NOIRLab Galaxy Group Talk
How to connect galaxies across cosmic time? Tucson, Jan. 2024

- **Carnegie arXiv Tea (invited)**
Relating galaxies across different redshift to study galaxy evolution Pasadena, Jan. 2024
- **KIPAC tea talk at Stanford University (invited)**
Characterizing the assembly of dark matter halos with protohalo size histories Stanford, Jan. 2024
- **UC Santa Cruz CGI (Cosmology/Galaxies/IGM) Seminar (invited)**
Central Galaxy Quenching and its Relation to Halo Formation Time & Large-scale Environment Santa Cruz, Jan. 2024
- **Galread: Princeton/IAS Galaxy Journal Club**
Characterizing the assembly of dark matter halos with protohalo size histories Remote, Oct. 2023
- **UC Santa Cruz CGI (Cosmology/Galaxies/IGM) Seminar**
Characterizing the assembly of dark matter halos with protohalo size histories Remote, Oct. 2023
- **KIAA-DoA Seminar, Peking University (invited)**
Central Galaxy Quenching and its Relation to Halo Formation Time & Large-scale Environment Beijing, Mar. 2023
- **Lunch Talk at the Department of Astronomy, Tsinghua University (invited)**
Relating galaxies across different redshift Beijing, Nov. 2022
- **Lunch Talk at Kavli-IPMU, University of Tokyo (invited)**
Finding proto-clusters to trace galaxy evolution Remote, Jun. 2021

CONFERENCE TALKS

- **DEX-XXII: 22nd Durham-Edinburgh eXtragalactic Workshop**
An EAGLE's view of the galaxy size evolution across cosmic time Edinburgh, Jan. 2026
- **SWIFTCON 2025**
Dissecting the scatter of stellar mass-halo mass relation in COLIBRE Leiden, Oct. 2025
- **L-GALAXIES workshop 2025**
GALFORM++: Next-generation semi-analytical galaxy formation model Berlin, Oct. 2025
- **National Astronomy Meeting (NAM)**
Contributed talks: Durham, Jul. 2025
 - ◆ Local Group Analogs in a cosmological context: Relating the velocity structure to the cosmic web
 - ◆ Testing galaxy formation models with the stellar mass-halo mass relations for star-forming and quiescent galaxies
 - ◆ Environmental Dependence of the Mass–Metallicity Relation in Cosmological Hydrodynamical Simulations**Poster:**
 - ◆ Dissecting two-halo galactic conformity effect for central galaxies
- **Expanding the boundaries of dark matter halo**
Local Group Analogs in a cosmological context Shanghai, May 2025
- **Mock Barcelona 2024 (invited)**
Stellar mass-halo mass relation to the second order Barcelona, Oct. 2024
- **Conference of the Co-evolution of galactic eco-systems and their large-scale environments (invited)**
Dissecting two-halo galactic conformity effect for central galaxies Hangzhou, Apr. 2024
- **The 2nd Shanghai Assembly on Cosmology and Structure Formation**
Characterizing the assembly of dark matter halos with protohalo size histories Shanghai, Oct. 2023
- **Collaboration Workshop on Cosmology and Galaxy Formation**
Relating Galaxies across Cosmic Time to study galaxy evolution Shanghai, Jun. 2023
- **25th Chinese Astronomical Society Guoshoujing Symposium on Galaxies and Cosmology**
Central Galaxy Quenching and its Relation to Halo Formation Time & Large-scale Environment Huangshan, May 2023

- Conference of Star Formation and Nuclei Activity in Galaxies
Central Galaxy Quenching and its Relation to Halo Formation Time & Large-scale Environment Nanjing, Mar. 2023
- The 11-th Prime Focus Spectrograph collaboration meeting
Identifying galaxy groups from high-z and incomplete spectroscopic surveys Pasadena, Dec. 2019
- The 10-th Prime Focus Spectrograph collaboration meeting
Finding groups/clusters of galaxies in the PFS galaxy evolution survey Shanghai, Dec. 2018